

Sensitivity Analysis: Pricing the Hidden Confounder

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 4, Hour 2 (slides 136 and 204). Same story as Day 3: a tutoring programme at one school.

Step 1. Where we are

Day 3 adjusted for prior grade X and found tutoring adds **+9** points. That rests on the bet that X is the only confounder. Sensitivity analysis does not test the bet; it prices it: **how strong would a hidden confounder have to be to erase the +9?**

Step 2. How a hidden confounder biases the estimate

Call the hidden variable U (say, a parent who pushes: pushes the child into tutoring and pushes them to study). U biases the adjusted estimate by two things multiplied:

$\text{bias} \approx (\text{share with U among tutored} - \text{share with U among untutored}) \times (\text{effect of U on the score})$
 $\text{bias} \approx \text{imbalance} \times \text{strength}$

The estimate is wrong by the bias. The true effect would be $9 - \text{bias}$. The +9 is erased when bias reaches 9.

Step 3. The tipping table

Each cell is the bias a hidden U would cause. Shaded cells erase the effect.

Imbalance ↓ / Strength →	U adds 5 points	U adds 10 points	U adds 15 points	U adds 20 points
0.2 (20 points more common among tutored)	1	2	3	4
0.4	2	4	6	8
0.6	3	6	9	12
0.8	4	8	12	16

To erase +9, U needs $\text{imbalance} \times \text{strength} \geq 9$: for example 60 percent more common among tutored students **and** worth 15 points on its own.

Step 4. Benchmark against what we can see

Is a confounder that strong plausible? Compare it with the confounder we measured, prior grade X:

imbalance of X: 3/4 of tutored are Low, 1/4 of untutored are Low → 0.50
strength of X: untutored High score 80, untutored Low score 52 → 28 points
product: $0.50 \times 28 = 14$

Prior grade, the strongest thing we know about a student, has a product of 14. A hidden U would need a product of 9, about two thirds as strong as prior grade, **while being independent of prior grade**, since we already adjusted for that. Possible, not likely. That sentence is the deliverable (slide 136).

Step 5. The same idea with the standard tools

Tool (slide 204)	Reports	Reading here
Benchmark table (this handout, Cinelli and Hazlett style)	bias for each imbalance $\times$ strength; compare with observed X	U must be two thirds of prior grade
E-value	one number: the minimum association U needs with both T and Y, on a risk-ratio scale	larger is safer; report it with the estimate
Rosenbaum bounds	for matched data: how much hidden bias in the odds of being tutored flips the p-value	$\Gamma = 1.5$ means a hidden U could make one student 1.5 times as likely to be tutored as their match

Three things to remember

1. Unconfoundedness cannot be tested, only priced.
2. Bias is imbalance times strength; both must be large to matter.
3. Always benchmark: "as strong as prior grade" means something; "strong" does not.